

MAJOR CONTRIBUTORS TO THIS REPORT



Anuj Deshpande,
founder, Makerville
Solutions



Chris Allsup,
technical marketing
manager, senior staff,
RTL synthesis and
test automation,
Synopsys Inc.



**Marco-Casale
Rossi**,
product marketing
manager, design group,
Synopsys Inc.



Neel Desai,
product marketing
manager, Lynx,
strategic alliances
and GTS marketing,
Synopsys Inc.



Ruchir Dixit,
technical director,
Mentor Graphics



**Sreekanth
Tamanna**,
country manager
for India, Pakistan,
Bangladesh and Sri
Lanka, Altium Ltd



Tom Ferry,
vice president -
marketing, silicon
engineering group,
Synopsys Inc.

has been working in this direction as well. Deepak says, "In the last year, we have focused on making our tools more user-friendly."

With the large number of available components and minute changes from one component to the next, accurate documentation is a necessity. Hence, "Efficient documentation tools are becoming a necessity," says Tamanna. Updated tools have automatic schematic and PCB generation, which reduces errors due to human intervention.

Taking on the challenges

Certain challenges can occur during designing of the circuit, mask preparation, testing the circuit and so on. For circuit designing, Rossi explains, "Once the objective is within reach, we hold placement and routing, and systematically change the gates—same footprint, different timing, power, temperature inversion point."

Routing the circuits. Nowadays, it is practically impossible to do custom/manual routing and, hence, designers use EDA tools to do auto-routing. According to Cadence design team, routing can take anywhere from a few hours to days, depending on the complexity of the circuit. For such complex SoCs that have billions of transistors, extracting resistance and capacitance is a Herculean task. Commercially-available field solvers do things automatically once placement and routing is done during design phase.

Masking. Primary challenge in

mask creation is for semiconductor manufacturers to continue scaling at and below 16nm/14nm using 193nm ArFi lithography equipment. "Until next-generation lithography equipment is available for production, manufacturers must find techniques to print smaller and smaller geometries with existing equipment," says Tom Ferry, vice president - marketing, silicon engineering group, Synopsys Inc.

He adds, "Computational techniques modify the design layout to create mask designs that enable lithography equipment to produce patterns on masks that accurately represent the layout (designer intent)."

Verification and synthesis. For many years, designers exclusively relied on stuck-at automatic test pattern generation to test ICs. Because delay faults caused by opens and shorts occur more frequently than stuck-at faults, they gradually began to employ transition delay automatic test pattern generation to improve defect coverage and lower defective parts per million.

According to Chris Allsup, technical marketing manager, senior staff, RTL synthesis and test automation, Synopsys Inc., "At advanced nodes, process variations give rise to increasingly-subtle physical defects that require additional patterns that target more complex failure mechanisms such as partial shorts and partial opens."

The RTL netlist in Verilog format is now accompanied by a power

intent description in IEEE 1801 UPF format, and logic synthesis has been upgraded to understand power and ground, and to perform optimisations, taking into account increasingly-complex concepts such as power/voltage/shutdown islands.

Database issues. Compatibility among EDA tools is another parameter to be considered. While files from one tool should be able to work with other systems for designers to have maximum ease of transition, the tools have to be lightweight in order to work efficiently. Rossi puts to rest the database debate with, "There are no simple databases in EDA, but rather efficient binary dumps on disks of in-memory data representation/infrastructure of different classes of tools."

Where the evolution is going

"Today, there is no solution for continuity between our logic synthesis and our place-and-route tools," adds Rossi. "Moving forward, the border between logic synthesis and place-and-route will blur, and possibly disappear."

Chip design is a constant battle against rising process and functional complexity on the one hand, and reducing time to market, on the other. Additions like the 5G library for system design, according to Deepak, helps innovation in technology, as well. We can hopefully look forward to a design environment that includes validation of IPs, libraries and foundry technology data at the beginning of the process. **EFY**